

AI TRACEFINDER

Document Source Identification using Scanner Pattern

Milestone – 3 Report

Infosys Springboard Internship

Submitted by:

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1. Introduction

With the rapid growth of digital documentation, scanned images are widely used in official, academic, and legal applications. Identifying the source scanner of a document plays a critical role in forensic analysis and authenticity verification. Scanner devices introduce unique sensor noise patterns that can be learned using advanced machine learning techniques.

Milestone-3 focuses on applying **Deep Learning techniques** to classify scanner devices using **PRNU-based images** and interpreting the model's decisions using **explainable AI methods**. This milestone builds upon the preprocessed and feature-extracted data prepared in earlier milestones.

2.Objective of Milestone-3

The objectives of Milestone-3 are:

- To design and train a Convolutional Neural Network (CNN) for scanner identification.
- To tune model hyperparameters and analyze training performance.
- To evaluate the trained model using standard classification metrics.
- To apply explainability techniques to understand model predictions.

3. Dataset Description:

The dataset used in this milestone consists of PRNU-based images extracted from scanned documents obtained from multiple scanner devices. These images represent scanner-specific noise characteristics that are independent of document content.

The dataset includes PRNU images corresponding to the following scanner models:

- Canon
- Epson
- HP

To ensure reliable evaluation, the dataset was divided into:

- Training set
- Validation set
- Test set

This separation avoids data leakage and enables fair performance assessment.

4. Deep Learning Model Development (Week-5)

4.1 CNN Architecture Design

A Convolutional Neural Network was designed to learn scanner-specific PRNU patterns. The architecture includes:

- Convolutional layers for feature extraction
- Batch Normalization layers for training stability
- Max Pooling layers for dimensionality reduction
- Fully Connected layers for classification
- Softmax output layer for multi-class prediction

The model was implemented using TensorFlow and Keras.

4.2 Model Training and Validation Performance

The Convolutional Neural Network was trained using PRNU-based images while monitoring both training and validation performance. Validation accuracy was used as a key indicator to evaluate the model's generalization capability and to avoid overfitting. During training, the model showed consistent improvement in learning scanner-specific PRNU features. The final validation accuracy achieved by the CNN model was:

Validation Accuracy: 96.34%

The training and validation accuracy curves indicate stable convergence of the model.

4.3 Hyperparameter Tuning

Hyperparameters such as:

- Learning rate
- Batch size
- Number of epochs

were adjusted to improve model generalization and stability. Fine-tuning was performed using a lower learning rate to refine learned features.

4.4 Data Augmentation Consideration

Since PRNU features are sensitive to geometric and photometric changes, aggressive data augmentation (such as rotation or heavy brightness changes) was avoided. This ensures preservation of scanner-specific noise patterns, which is essential for forensic analysis.

5. Model Evaluation (Week-6)

5.1 Evaluation Metrics

The trained CNN model was evaluated using the test dataset. The following metrics were computed:

- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix

These metrics provide a comprehensive assessment of the model's classification performance across different scanner classes.

5.2 Performance Analysis

The trained CNN model was evaluated on the validation and test datasets using standard classification metrics. The results demonstrate that the model effectively learns scanner-specific noise patterns.

The key performance results are summarized below:

- **Training Accuracy:** 97.28%
- **Validation Accuracy:** 96.34%
- **Test Accuracy:** 93.08%

The validation accuracy reflects the model's ability to generalize to unseen PRNU images. Performance variations are primarily influenced by dataset size and scanner class distribution.

6. Model Explainability using Grad-CAM

6.1 Explainability Objective

To interpret the CNN's predictions, Grad-CAM (Gradient-weighted Class Activation Mapping) was applied. Grad-CAM helps visualize the regions of the input image that contribute most to the model's decision.

6.2 Grad-CAM Observations

The generated Grad-CAM heatmaps highlight noise-like regions distributed across the image rather than document text or layout. This confirms that the CNN relies on scanner-specific PRNU characteristics, validating the forensic reliability of the model.

7. Outcome of Milestone-3

By the completion of Milestone-3, the following outcomes were achieved:

- CNN model trained successfully using PRNU-based images
- **Validation accuracy of 96.34% achieved**, demonstrating effective generalization
- Hyperparameters tuned for improved stability
- Model evaluated using accuracy, F1-score, and confusion matrix
- Explainability achieved through Grad-CAM visualization

8. Conclusion

Milestone-3 successfully demonstrated the application of deep learning techniques for scanner identification using PRNU features. The CNN model effectively learned scanner-specific patterns and achieved reliable classification performance. The integration of Grad-CAM provided transparency and interpretability, ensuring that the model's decisions are based on meaningful forensic features. This milestone completes the core modeling and evaluation phase of the AI TraceFinder project.

9.Conclusion

Milestone-2 successfully demonstrated the effectiveness of PRNU-based correlation for scanner identification. While baseline machine learning models achieved moderate accuracy, the PRNU-correlation method significantly improved performance by leveraging sensor-specific noise fingerprints.

This milestone establishes a strong analytical foundation for further exploration of deep learning-based approaches in subsequent milestones.